MIL-T-36487B
12 June 1978
SUPERSEDING
MIL-T-36487 (DSA-DM)
25 October 1966 and
MIL-T-0036487A (DSA-DM)
3 July 1972

MILITARY SPECIFICATION

TUBE, CAPILLARY, BLOOD SAMPLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain and heparin-coated capillary tubes for use in microhematocrit procedures.

1.2 Classification.

1.2.1 Types. Capillary tubes shall be of the following types, as specified for the appropriate stock number and item identification (see 6.2):

Type I - Heparinized
Type II - Plain

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Set-Up.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors, in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Glass shall be soda-lime or borosilicate glass, suitable for the intended purpose.

3.1.2 Heparin. Heparin shall be the ammonium salt isolated from lungs or the intestinal mucosa of beef or pork.

3.2 Construction. Capillary tubes shall be fabricated of glass, and shall be straight and open at both ends without lip or constriction. Bore of the capillary tube shall be uniform, and taper shall not exceed 0.001 inch (in.) in 3 in. Tube ends shall be cut at 90 degrees to tube axis, and shall not have sharp, chipped or jagged ends. The capillary tubes shall have the following dimensions (in millimeters):

Inside diameter	1.07 to 1.24
Length	75 ± 2
Wall thickness	0.20 +.03 -.02

3.2.1 Type I. Inner surfaces of type I tube shall be completely coated with ammonium heparin. Tube shall contain a minimum of 3.0 U.S.P. units, when tested as specified(see 4.5.1). Before the coating process is applied, the tubes shall be clean and dry, and free of foreign material.

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3.2.2 Type II. Type II tubes shall be plain and free of foreign material. Tubes shall be clean and dry, and ready for use.

* 3.3 Color coding. Each capillary tube shall be color coded. The type I tube shall have a red tip or band and the type II tube shall have a blue tip or band.

3.4 Performance. The inner surfaces, except as specified, shall be clean and free of filmy material which may impair the performance (see 3.4.1 and 3.4.2) of the tube.

* 3.4.1 Capillarity. The capillary tube shall be capable of drawing up blood or sheep plasma to a level within 2 centimeters (cm) from the top end when tested as specified (see 4.5.2).

3.4.2 Fluidity (For type I only). No coagulation of the blood or sheep plasma shall be evident when tested as specified (see 4.5.3).

* 3.5 Workmanship. Workmanship shall be first class throughout. The capillary tubes shall be free from defects which detract from their appearance or impair their serviceability. The bore of the capillary tube shall be free of constrictions or debris that interfere with drawing up blood.

3.6 Lot or control number (For type I only). A lot or control number shall be indicated on the interior packages and exterior containers (see 5.3) which can be traceable to specific production run(s) involving the same batch of ammonium heparin.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.2 Sampling.

* 4.2.1 For examination. Sampling for examintaion shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for inspection purposes shall be the individual capillary tube. No more than five capillary tubes shall be selected from any one unit package (see 5.1.1). For count, the unit of product for sampling purposes shall be the unit package.

TABLE I. For examination.

	Inspection level	AQL percent defective
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5
For taper examination	S-3	1.0
For count	S-1	*

* Acceptance number shall be zero (see 4.3.4).

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The acceptance number shall be zero. The unit of product for testing of capillary tubes shall be the unit package. The fluidity test shall be run on each lot of ammonium heparin material for the type I capillary tubes.

TABLE II. For tests.

Characteristic	Requirement	Test procedure	Inspection level
Heparin content	3.2.1	4.5.1	S-4 *
Capillarity	3.4.1	4.5.2	S-1 **
Fluidity	3.4.2	4.5.3	S-1 **

* For purposes of this test, 10 packages shall be combined into one test sample. If the number of packages from the sampling plan is not divisible by 10, a sufficient number of packages shall be added to the odd packages to make a test sample.

** One tube shall be selected from a unit package. No more than one unit package may be selected from an intermediate package (see 5.1.2).

4.3 Examination. The tubes shall be examined to determine compliance with all requirements contained in this specification.

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* 4.3.1 Classsification of defects. Examination shall be conducted in accordance with table III classification of defects.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Tube not free of cracks or breaks.
102	Tube not free of sharp, chipped, or jagged ends.
103	Inside surface not completely coated (type I only).
104	Tube not clean or dry, or free of foreign materials.
105	Tube not cut at 90 degrees to the axis of the tube.
106	Tube not straight.
Minor	
201	Color code on tube not complete.
202	Glass not free of scratch marks.

* Examination shall not be restricted to the classified possible defects listed above.

* 4.3.2 Dimensional examination. The tubes shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be considered as a defect.

* 4.3.3 Taper examination. The tubes shall be examined for defects in taper of the bore. Any taper not within the tolerance specified herein shall be considered as a defect.

4.3.4 Count. The number of tubes in the unit package shall be determined and shall be not less than 95 tubes for an individual package; the average count shall be not less than 100 tubes per unit package. The number of tubes in the unit package shall be determined by weighing 100 tubes, and calculating the number per sample package based on weighing the total quantity in a sample package.

* 4.4 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.5.1 Heparin content. Heparin content shall be determined by the method of assaying sodium heparin in the latest edition of the U.S.P. The results obtained shall represent the heparin content on the inner surfaces of the tubes. No heparin from the outside surface shall enter the test sample.

* 4.5.2 Capillarity test. Held in a horizontal position, the sample capillary tubes shall be tested in accordance with currently accepted technique for hematocrit determination. The level to which the blood is drawn up within a 30 second period shall be noted.

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Sheep plasma may be used in lieu of blood and shall be prepared in accordance with U.S.P. Assay for Sodium Heparin.

* 4.5.3 Fluidity test. The test shall be conducted by initially preparing recalcified sheep plasma by the following two step process:

(a) Sheep plasma preparation. Sheep plasma shall be prepared in accordance with U.S.P. Assay for Sodium Heparin.

(b) Recalcify sheep plasma. To 10 milliliters (ml) of the prepared sheep plasma shall be added 2.0 ml of the 1.0% calcium chloride solution used in the heparin assay (4.5.1). The sheep plasma and calcium chloride solution shall be mixed well.

In addition, samples of both the plain sheep plasma and recalcified sheep plasma shall be used as controls in accordance with the following:

(a) Positive control. A plain (i.e. non-heparinized) capillary tube shall be filled with recalcified sheep plasma.

(b) Negative control. A heparinized capillary tube shall be filled with plain sheep plasma (see 4.5.3 a).

Immediately after the preparation of the recalcified sheep plasma, the capillary tubes shall be filled by immersing the tips in the recalcified sheep plasma and holding the tubes at such an angle as to facilitate quick filling. The capillary tubes shall be filled to 3 mm from the other end. At the end of a one-hour period, the sample tubes shall be inspected for evidence of coagulation by carefully snapping off segments of tubing which shall be placed flat on a surface of black background to facilitate observation and comparing with the control samples. Evidence of coagulation occurring is indicated by the sheep plasma becoming opaque or the presence of a fine fibrin thread.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package, containing 100 tubes as specified, constitutes one unit of issue.

* 5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in the procurement document.

TABLE IV. Packaging quantities.

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

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5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

* 5.1.3.1 Unit package. Each unit shall be packaged in a plastic vial or paperboard box of suitable size and design. Capillary tubes shall be suitably packaged to prevent breakage. Closure shall be adequate to prevent entrance of dust or spilling of contents.

* 5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676 or PPP-B-636, type CF, class domestic, except that commercial colors shall be acceptable. Closure shall be as specified in the applicable box specification.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

* 5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure, waterproofing and strapping shall be as specified in the box specification.

5.2.2 Level B.

* 5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

* 5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

* 5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. The lot (control) number and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15^{0} - 30^{0}C (59^{0} - 86^{0}F)" shall also be shown for type I capillary tubes.

* 5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. The lot (control) number and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15^{0} - 30^{0}C (59^{0} - 86^{0}F)" shall also be shown for type I capillary tubes.

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6. NOTES

6.1 Intended use. Capillary tubes are intended for use with the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-299-9835	Centrifuge, Laboratory, Microhematocrit. 115 Volt, 60 Hz, AC
6630-00-585-1378	Reader, Microhematocrit, Spiral type.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of tube required (see 1.2.1).
(c) National Stock Number (NSN).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is applicable to the following items listed in the Federal Supply Catalog:

National Stock Number	Item Identification	Specification designated type
6630-00-618-0072	TUBE, CAPILLARY, BLOOD SAMPLE, Glass, Heparinized, 100s	I
6630-00-618-0073	TUBE, CAPILLARY, BLOOD SAMPLE, Glass, 100s	II

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.5 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
Defense Logistics Agency - DM

Project No. 6630-0292

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

TUBE, CAPILLARY, BLOOD SAMPLE MIL-T-36487B

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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